ZBE TOLL ON STELLAR · HACKMERIDIAN 2026

No fine. A toll you can appeal.



An opt-in, appealable toll for Low Emission Zones.

Team IMPEDANCIA UNO · Genesis track

THE PROBLEM

Low Emission Zones are spreading. The way they’re enforced isn’t working.

1

Enter without a valid label

Access depends on the car’s environmental label: Zero, Eco, C, B or none.

2

A fine arrives weeks later

By post. The driver had no way to just pay and go.

3

The city pays to collect

A slow, expensive procedure to collect a small amount.

THE IDEA

Turn the fine into a fair, voluntary toll on Stellar.

FOR DRIVERS

who would rather pay a daily fee than get a fine

They top up a prepaid **ZBE pass**. If they enter without a valid label, the day's fee is locked — and they can appeal it.

FOR CITIES

that want a fair, cheap and verifiable procedure

Zone rules are public on-chain, fees settle straight to the treasury, and disputes go to an independent arbiter.

FOUR PRINCIPLES

Not an automatic fine.

Consent first

The pass is opt-in. Without one, nothing is ever charged on-chain.

Locked, not taken

The fee is locked in the contract. The city can’t touch it before the deadline.

Always appealable

An independent arbiter decides — never the party that collects.

Private by design

Only an HMAC-SHA256 of the plate goes on-chain, never the plate.

HOW AN ENTRY WORKS

From camera to treasury, on-chain.

01

Camera reports

An authorized camera signs the plate's HMAC and the zone.

02

Rules check

Label allowed: free. No pass: only a **no_bono** event, nothing charged.

03

Fee locked

Moves from the pass's available to its locked balance. Status: Pending.

04

Window closes

No appeal: anyone settles it to the treasury. Appeal: the arbiter decides.

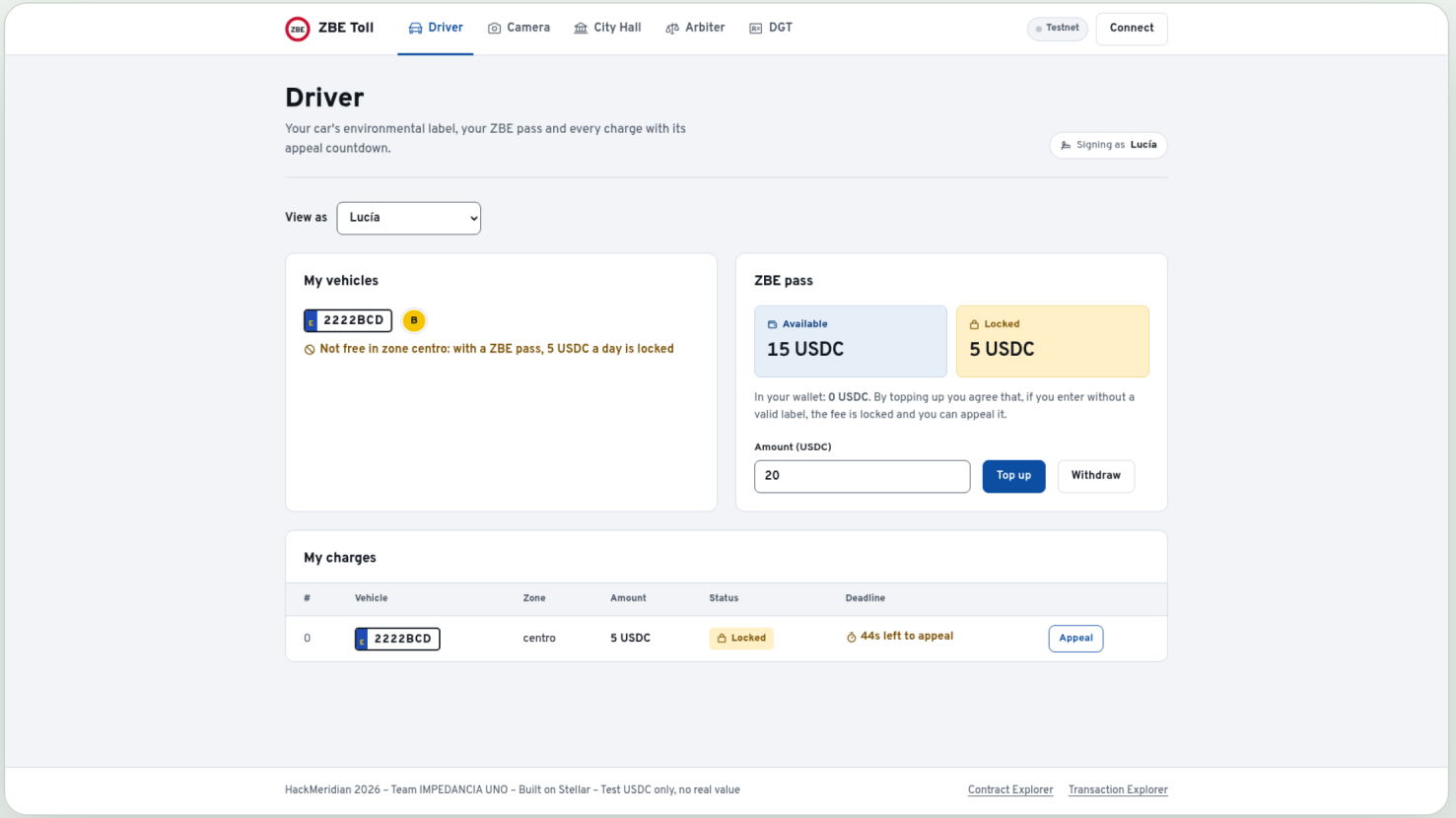
One charge per vehicle, zone and day — like London's ULEZ daily charge.

LIVE ON TESTNET

5 USDC locked, not taken.

Lucía's car has a B label. It enters zone *centro*, which only admits Zero and Eco.

Her pass locks the day's fee and a countdown to appeal starts.

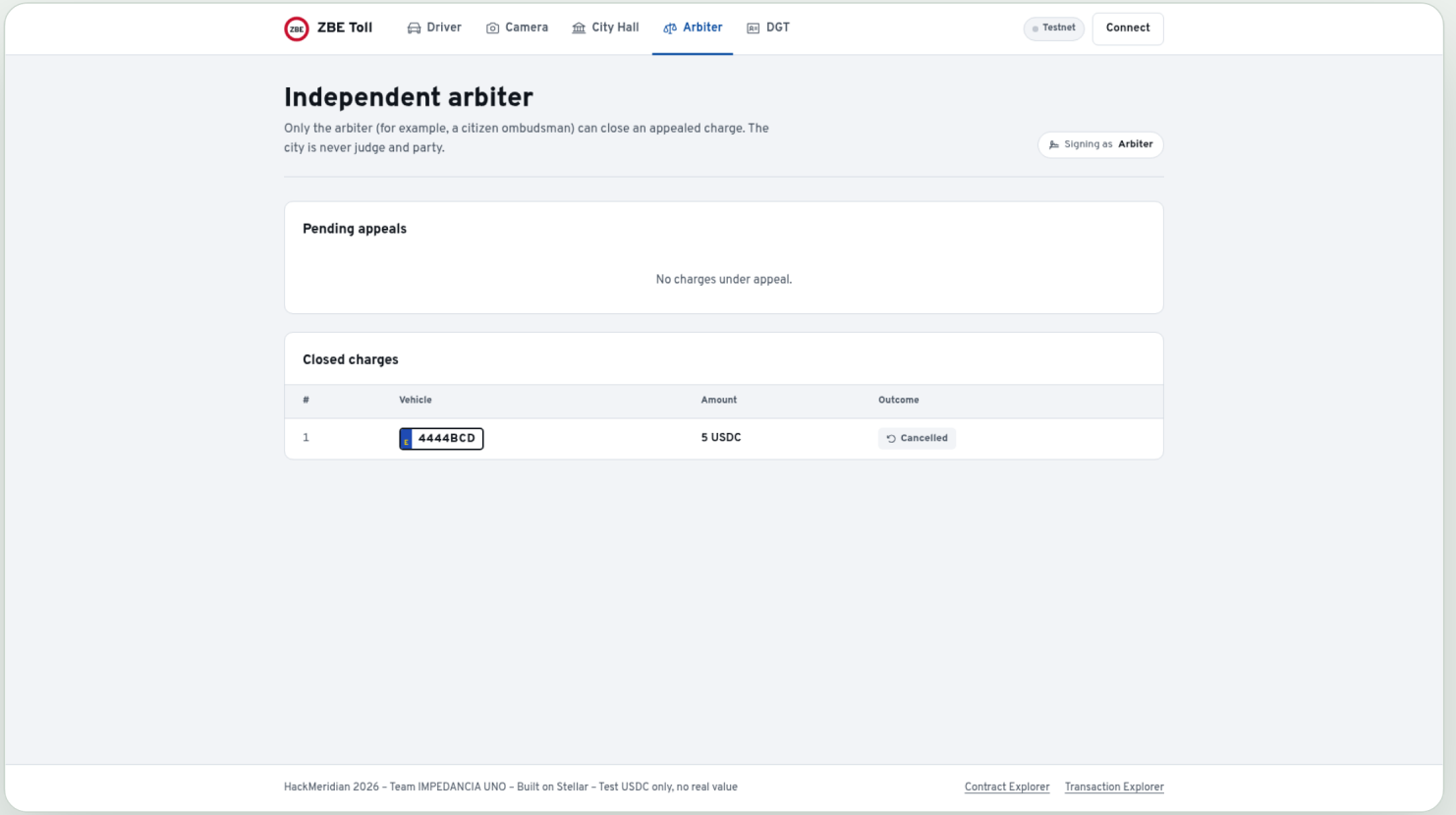


LIVE ON TESTNET

The collector never judges.

No appeal: when the window closes, City Hall settles the fee to the treasury.

Appeal: only the independent arbiter can uphold or cancel. Cancelled fees return to the pass.



WHY STELLAR

Rules everyone can verify. Money nobody can grab early.

Labels as tokens

Non-transferable. Only the DGT can issue, reassign or revoke them.

Public zone rules

Allowed labels, fee and appeal window live on-chain.

Escrow in a contract

The city can’t touch a locked fee before the deadline.

Stablecoins and Soroban

Cheap, fast payments in USDC/EURC; the whole procedure as smart contracts.

ARCHITECTURE

No backend. Two contracts.

etiqueta

Label contract

Admin: DGT. issue, reassign, revoke. No transfer for the owner.

peaje

Toll contract

Admin: City Hall. Zones, cameras, passes, charges, appeals, settlement. Reads labels from etiqueta.

USDC

Stellar Asset Contract

Holds the passes; pays the treasury.

Every actor is a view of the web app that signs with its own key:

Driver · Camera · City Hall · Arbiter · DGT

WHAT WE BUILT

A working prototype on Stellar testnet.

- **Two Soroban contracts** in Rust, tested with a Stellar Asset Contract token.
- **Published on Stellar Registry** and deployed on testnet.
- **React web app** (Stellar Scaffold) with five role views and Freighter.
- **Full flow end to end:** top up, lock, appeal, resolve, settle.

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unverified/zbe-etiqueta-impedancia-uno
unverified/zbe-peaje-impedancia-uno
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NEXT STEPS

From prototype to a city pilot.

Pilot with a city

Real cameras signing their own reports.

DGT integration

Labels issued from the official registry.

More zones

Several zones and cities with their own rules and fees.

Lower storage cost

Measure rent per charge on mainnet and optimize it.

ZBE TOLL ON STELLAR

“Not an automatic fine: a toll you choose, you can appeal, and nobody can collect early.”

Team IMPEDANCIA UNO

HackMeridian 2026 · Genesis track

github.com/marcocpp/peaje-zbe

Demo video: youtube.com/watch?v=ys-ihhcaj7s